SQL i zapytania SQL SELECT
SQL and SQL SELECT queries

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The role of SQL

- SQL – Structured Query Language (*Strukturalny język zapytań*)– the language used to:
  - Query the data i.e. get raw or aggregated data from one or many tables at the same time
  - Define data structures e.g. create new tables
  - Update data in a database by:
    - Creating new records
    - Modifying records
    - Deleting records
- The SQL language is used by:
  - Software developers as software developers send SQL queries from client applications to read and modify database content
  - Data scientists to retrieve data from a database to use it for further steps of analytical tasks
  - Database administrators to manage database content

SQL revisions

- Different revisions of the standard exist, namely:
  - SQL:2023
  - SQL:2016
  - SQL:2011
  - SQL:2008
  - SQL:2003
  - SQL99 a.k.a. SQL3
  - SQL92 a.k.a. SQL2
  - SQL86 a.k.a. SQL1 developed by ANSI (American National Standards Institute) and ISO (International Standards Organisation)
- As an example, SQL:2023 includes extensions supporting graph-like queries in databases, and improved support for JSON

Standards including 2023 version can be purchased at https://www.iso.org

SQL implementations

- SQL standards:
  - Initially based on the common set of features implemented in IBM and Oracle,
  - Only partly implemented by some DBMS vendors,
  - Thus, writing SQL code that is portable among different DBMSs is not easy
  - Different DBMS vendors offer extensions to the language that include procedural statements like PL/SQL by Oracle, Transact-SQL by Microsoft
- Already SQL3 defined:
  - Object-oriented extensions,
  - Core specification that should be implemented by every vendor of RDBMS,
  - Optional packages addressing the needs of data warehouses, spatial data, time-based data, multimedia and other fields.
- Due to the complexity of recent SQL revisions, the division into Core and Optional part is maintained
- Hence, typical SQL dialects are compared against Core SQL as found, e.g. in SQL:2023 Part 2 (Foundation) and Part 11 (Information and definition schemas)

DBMS – compliance with SQL standard – sample compliant subfeatures

Feature ID, Feature	Support
E011, Numeric data types	Oracle fully supports this feature.
E021, Character data types	Oracle fully supports these subfeatures: • E021-01, `CHARACTER` data type • E021-07, Character concatenation • E021-08, `UPPER` and `LOWER` functions • E021-09, `TRIM` function • E021-10, Implicit casting among character data types

These subfeatures of Core SQL standard are fully supported by Oracle 19c SQL dialect

Source: Oracle 19c SQL Language Reference
https://docs.oracle.com/en/database/oracle/oracle-database/19/sqlrf/Oracle-Compliance-To-Core-SQL2011.html#GUID-D372D906-805B-49B8-824A-D4697B05B7F8

DBMS – compliance with SQL standard – sample non-compliant subfeatures

Feature ID, Feature	Support
	Oracle has equivalent functionality for these subfeatures: • E021-04, `CHARACTER_LENGTH` function: use `LENGTH` function instead • E021-05, `OCTET_LENGTH` function: use `LENGTHB` function instead • E021-06, `SUBSTRING` function: use `SUBSTR` function instead • E021-11, `POSITION` function: use `INSTR` function instead

These subfeatures of Core SQL standard are handled by equivalent functionality by Oracle 19c SQL dialect

Source: Oracle 19c SQL Language Reference
https://docs.oracle.com/en/database/oracle/oracle-database/19/sqlrf/Oracle-Compliance-To-Core-SQL2011.html#GUID-D372D906-805B-49B8-824A-D4697B05B7F8

SQL SELECT

- SELECT statement serves to retrieve data from a table or a set of tables linked by relations (that is by foreign and primary keys)
- SQL SELECT statement can be executed also on the output of another SQL statement – treated as a pseudo-table

SQL SELECT – part I

type	building	floor	room
lecture	Main Building	2	213
lecture	Main Building	3	315
	. . .		. . .
lab	Chemistry	4	452

```
SELECT * FROM rooms
SELECT building, room FROM rooms
```

SELECT - aliases

```
SELECT building AS bld, room as rm
FROM rooms
```

The output of the query:

bld	rm
Main Building	213
Main Building	315
. . .	. . .
Chemistry	452

Lowercase and uppercase names can be used interchangeably in SQL keywords, such as „AS” here.

SELECT – WHERE clause

```
SELECT * FROM rooms where floor>2 and
type='lecture'
```

The output of the query:

type	building	floor	room
lecture	Main Building	3	315
	. . .		. . .

SELECT – IN phrase

```
SELECT * FROM rooms where floor IN (2,4)
```

The output of the query:

type	building	floor	room
lecture	Main Building	2	213
...	...		...
lab	Chemistry	4	452

The IN phrase can be used to specify a set values. Only rows with the column value in this set will be returned.
Note that performance may be reduced when the set includes several values e.g. the output of an embedded SQL SELECT query.

```
SELECT * FROM rooms where floor IN
(SELECT floor from rooms b where b.type='lecture')
```

SELECT – LIKE phrase

```
SELECT * FROM rooms where building LIKE
'Main%'
```

The output of the query:

type	building	floor	room
lecture	Main Building	2	213
…	. . .		. . .

The LIKE phrase can be used to specify a pattern to be matched by the values in a text column. Examples include:

- **% - any number of characters**
- **_ - any single character**
- **[b-g] – any character between b and g**
- **[ghj] – any of the listed characters**

Note that patterns should not be used too frequently, due to performance reasons

SELECT – ORDER BY clause

```
SELECT * FROM rooms where floor>=2 ORDER
BY type,floor
```

The output of the query:

type	building	floor	room
lab	Chemistry	4	452
lecture	Main Building	2	213
lecture	Main Building	3	315
	. . .		. . .

SELECT – ORDER BY clause

```
SELECT * FROM rooms [. . .] ORDER BY
Column_Name [DESC] [,Column_Name. . .]
```

- It is also possible to sort in a descending order
- Ascending order is default
- Please notice that text columns are sorted as text even if they contain numbers thus '1000'<'2'
- When multiple columns are used, rows that have the same values for the first column are compared in terms of the second column and so on

SELECT – DISTINCT clause

```
SELECT DISTINCT type, floor FROM rooms
where floor>=2
```

The output of the query:

type	floor
lab	4
lecture	2
lecture	3

Unique combinations of column values/expressions are obtained

Linking tables and Cartesian product
Złączanie tabel i iloczyn kartezjański

Rooms

type	Bld_code	Floor	Room
lecture	MAIN	2	213
lab	CHEMISTRY	4	452

Assets

Bld_code	Asset_ id	Room	Type
Main	5	213	Chair
Main	7	213	Table

How to perform cross-table query e.g. list of assets in main building on floor 2?

Linking tables –query styles

WHERE-BASED QUERY

```
SELECT … FROM
TABLE1,TABLE2[,…]
WHERE Condition1[ AND
Condition2 …] […]
```

JOIN-BASED QUERY

```
SELECT … FROM TABLE1
JOIN TABLE2 ON
Condition1 [JOIN … ON
…]
```

Example

```
SELECT * FROM rooms,
assets WHERE
rooms.bld_code=
assets.bld_code AND
rooms.room=assets.room
```

Example

```
SELECT * FROM rooms
JOIN assets ON
rooms.bld_code=
assets.bld_code AND
rooms.room=assets.room
```

Query interpretation – step I

Cartesian product is obtained

Bld_code	Floor	Room	Bld_code	Type	Room	…
MAIN	2	213	MAIN	Chair	213	…
CHEM	4	452	MAIN	Chair	213	…
MAIN	2	213	MAIN	Table	213	…
CHEM	4	452	MAIN	Table	213	…

To answer a query referring to multiple tables *Cartesian product* is created first i.e. a temporary listing of all possible combinations of records from the tables is created

Query interpretation – step II

Bld_code	Floor	Room	Bld_code	Type	Room	…
MAIN	2	213	MAIN	Chair	213	
CHEM	4	452	MAIN	Chair	213	
MAIN	2	213	MAIN	Table	213	
CHEM	4	452	MAIN	Table	213	

In the second step Cartesian product is filtered out so as to contain only rows satisfying the condition: `rooms.bld_code= assets.bld_code AND rooms.room=assets.room`

Cross-table queries

- The uniqueness of column names in the result set may be preserved by a DBMS. Thus, a suffix may be added to original column names e.g. bld_code_a, bld_code_b columns may be observed in the query output
- In case we need to relate to a column in the query:
  - If a column name appears only once in all the involved tables, it is enough to use unqualified name e.g. `floor`
  - Otherwise, fully qualified name should be used e.g. `rooms.bld_code`

Other JOIN categories

Rooms

Bld_code	Floor	Room
MAIN	2	213
CHEM	4	452

Assets

Bld_code	Type	Room	…
MAIN	Chair	213	…
MAIN	Table	213	…

How to obtain room 452 in the chemistry building in the output result set when there are no assets in the room?

Other JOIN categories

Rooms

Bld_code	Floor	Room
MAIN	2	213
CHEM	4	452

Assets

Bld_code	Asset	Room
MAIN	Chair	213
MAIN	Table	213

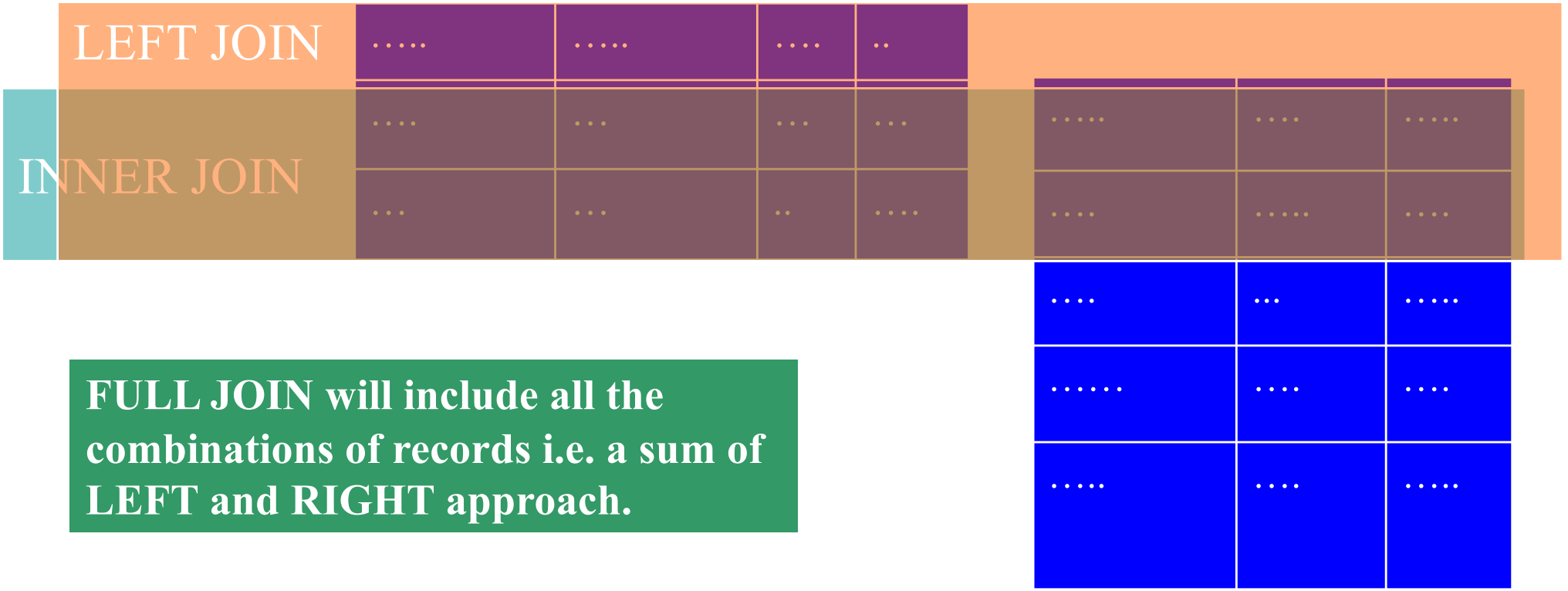
JOIN-BASED QUERY

```
SELECT … FROM TABLE1
[INNER|LEFT|RIGHT|FULL]
JOIN TABLE2 ON
Condition1[ AND
Condition2 …] […]
[ JOIN …]
```

Other JOIN categories

Table_1

Table_2



FULL JOIN will include all the combinations of records i.e. a sum of LEFT and RIGHT approach.

Join statements

- Most JOIN statements use primary key and foreign key to relate tables
- Example:
  - `SELECT * FROM Customers C`

```
JOIN Orders O ON
O.Customer_ID=C.Customer_ID
```

JOIN categories

Category	Description
INNER JOIN or simply JOIN	The only JOIN available in the old-style syntax. Only matching combinations of records in joined tables are left
LEFT JOIN	All the records from the parent table are left. In case they have no corresponding records in the joined table, NULL values are used for the columns of joined table.
FULL JOIN	All the records from the parent table and joined table are left. In case they have no corresponding records in the matching table, NULL values are used. Rarely used.
RIGHT JOIN	All the records from the joined table are left. In case they have no corresponding records in the parent table, NULL values are used. Rarely used.

Is there anything wrong here?

- Get a list of all customer names and the dates of their orders
- Solution?
  - This one:

    ```
    select CompanyName,orderdate from Customers C
    join orders O on o.CustomerID=c.CustomerID
    ```

  - Or:

    ```
    select CompanyName,orderdate from orders O join
    Customers C on o.CustomerID=c.CustomerID
    ```

Filtering the content of a table based on conditions referring to other tables

EXISTS phrase

- How to find all the records from the parent table that do [not] have matching records?
- Example:
  - All the customers that have not made any orders
    - `SELECT * FROM Customers C WHERE [NOT] EXISTS (SELECT * FROM Orders O WHERE O.Customer_id=C.Customer_id)`

EXISTS - algorithm

Rooms

Room_id	Bld_code	Floor	Room
7	MAIN	2	213
15	PHYSICS	4	452

Assets

Asset_id	Room_id	Type
5	15	Chair
7	15	Table
12	18	Projector

```
SELECT * FROM Rooms  R WHERE EXISTS
(SELECT * FROM Assets A  WHERE
R.room_id=A.room_id)
```

For every parent record (here: rooms record), a subquery is executed (here: `(SELECT * FROM Assets A  WHERE R.room_id=A.room_id)` and condition is checked (is there anything in the output of the query?)

EXISTS – step I

Rooms

Room_id	Bld_code	Floor	Room
7	MAIN	2	213
15	PHYSICS	4	452

Assets

Asset_id	Room_id	Type
5	15	Chair
7	15	Table
12	18	Projector

```
SELECT * FROM Rooms R WHERE EXISTS
(SELECT * FROM Assets A  WHERE R.room_id=A.room_id)
```

7	MAIN	2	213

```
 (SELECT * FROM Assets A  WHERE 7=A.room_id)
```

First record of rooms is processed. No assets with room_id=7 exist. Subquery above returns no rows. Hence, room 7 is not placed in the output.

EXISTS – example: step II

Rooms

Room_id	Bld_code	Floor	Room
7	MAIN	2	213
15	PHYSICS	4	452

Assets

Asset_id	Room_id	Type
5	15	Chair
7	15	Table
12	18	Projector

```
SELECT * FROM Rooms  R WHERE EXISTS
(SELECT * FROM Assets A  WHERE R.room_id=A.room_id)
```

15	PHYSICS	4	452

```
 (SELECT * FROM Assets A  WHERE 15=A.room_id)
```

Two assets with room_id=15 exist. Subquery above returns two rows. Hence, room 15 is placed in the output.

EXISTS – example: step III

Rooms

Room_id	Bld_code	Floor	Room
7	MAIN	2	213
15	PHYSICS	4	452

Assets

Asset_id	Room_id	Type
5	15	Chair
7	15	Table
12	18	Projector

```
SELECT * FROM Rooms  R WHERE EXISTS
(SELECT * FROM Assets A  WHERE R.room_id=A.room_id)
```

Room_id	Bld_code	Floor	Room
15	PHYSICS	4	452

In this case, the query returns only one out of two rows present in rooms.

Combining the results of multiple queries

UNION phrase

- Allows to add the output of multiple queries
- `SELECT …. FROM …..`

  `UNION [ALL]`

  `SELECT …. FROM ….`
- If ALL is skipped duplicate rows that can be found in multiple source queries are replaced with a single one

UNION - example

Assuming these two outputs of SQL queries are combined with UNION

Customer_ID	Order_Date	Order_Value	Delivery_City	
MAIN	01/02/2025	200.00	Warszawa	…

Customer_ID	Order_Date	Order_Value	Delivery_City	…
MAIN	02/03/2020	100.00	Poznań	
MAIN	14/11/2023	20	Kraków	….

Customer_ID	Order_Date	Order_Value	Delivery_City	…
MAIN	01/02/2025	200.00	Warszawa	…
MAIN	02/03/2020	100.00	Poznań	
MAIN	14/11/2023	20	Kraków	….

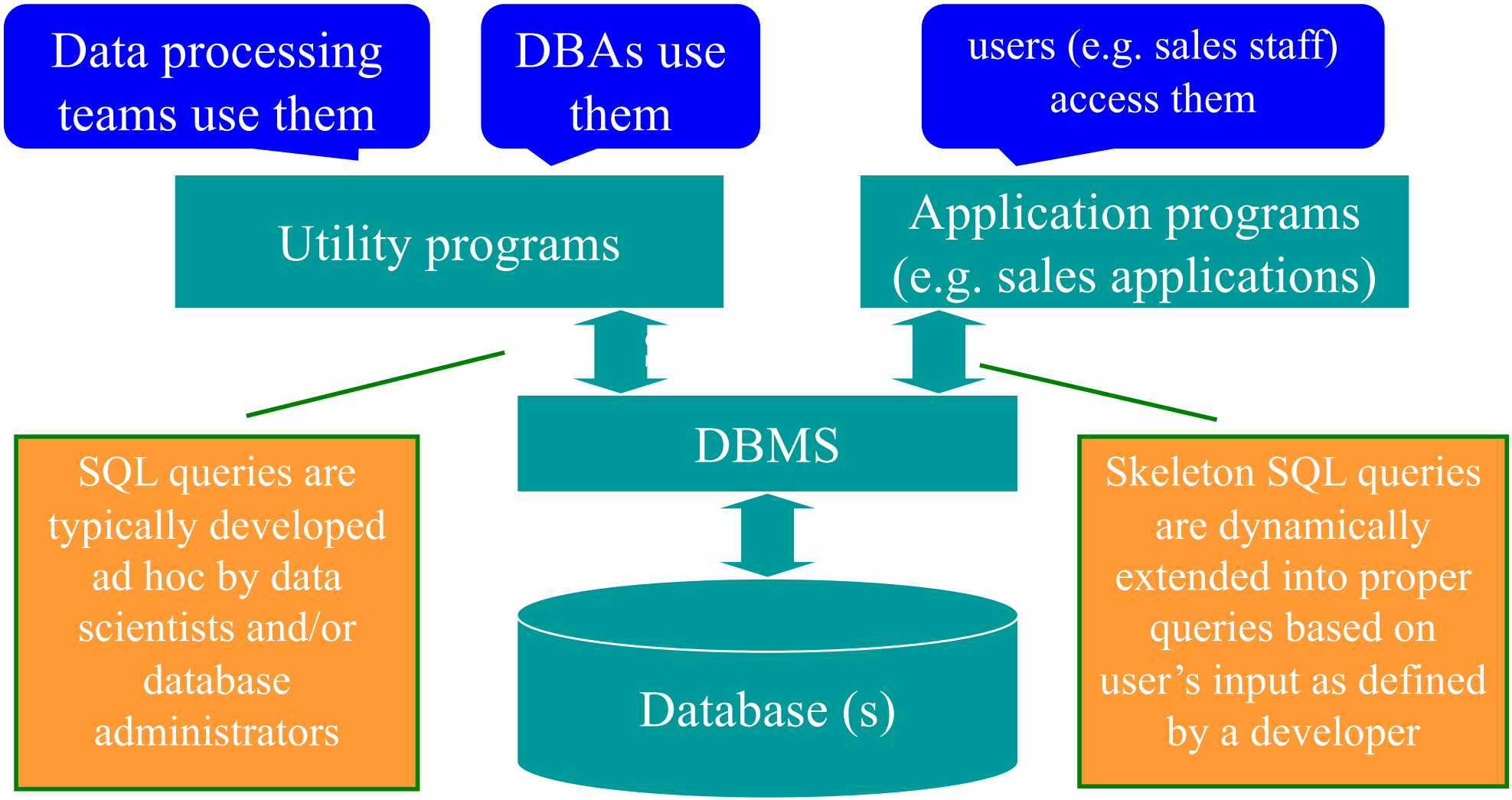
The output will be just all the rows from both input result sets

UNION - requirements

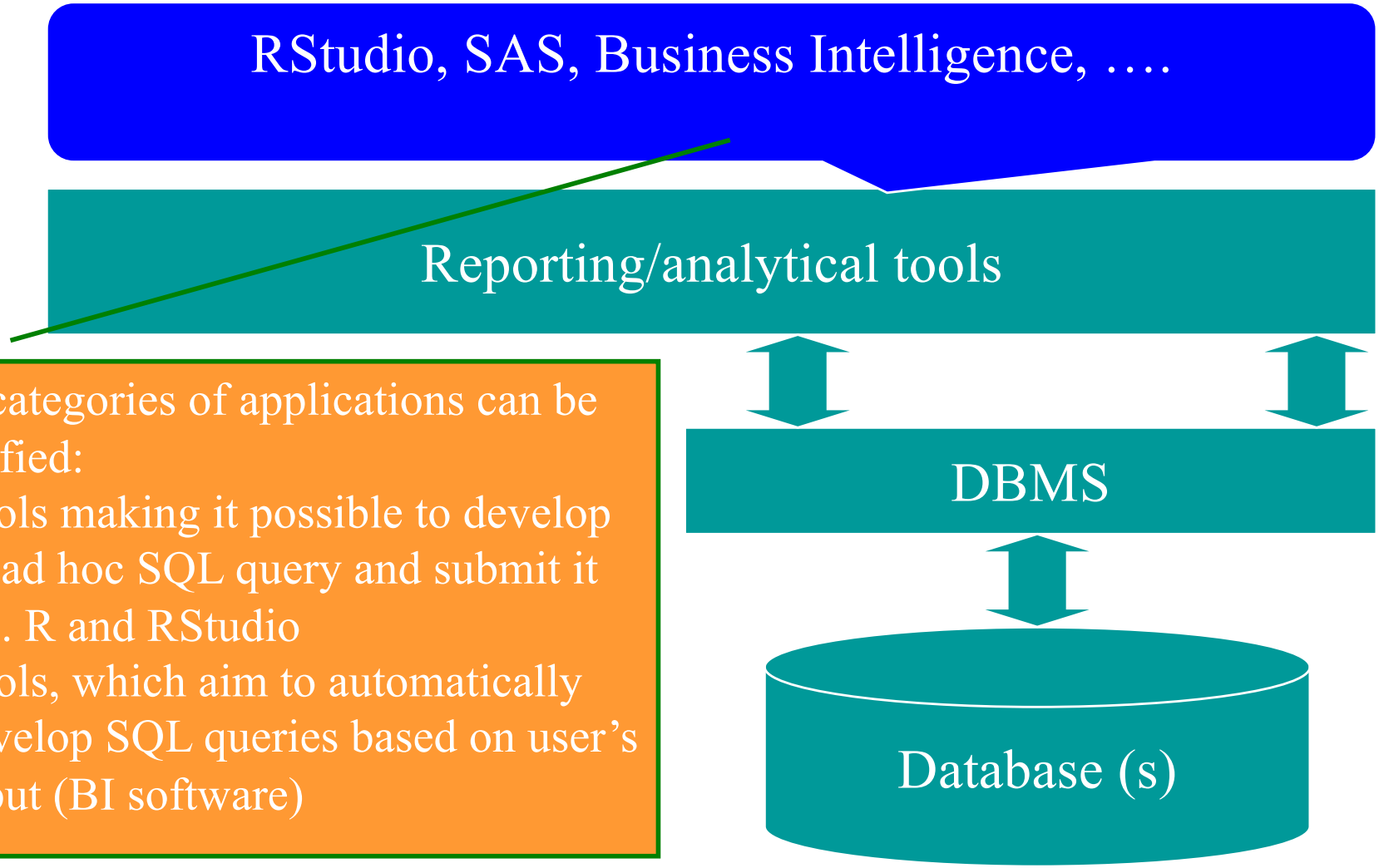
- The number of output columns must be the same in all participating subqueries
- The types of all the columns must be the same in all participating subqueries
- If a column value is defined by an expression, it may be necessary to declare its type to avoid ambiguous data type of that column

The role of SQL in database systems

DBMS and SQL


Data processing teams use them
DBAs use them
users (e.g. sales staff) access them
Utility programs
Application programs (e.g. sales applications)
DBMS
SQL queries are typically developed ad hoc by data scientists and/or database administrators
Database (s)
Skeleton SQL queries are dynamically extended into proper queries based on user's input as defined by a developer

SQL and analytical software



Two categories of applications can be identified:

- Tools making it possible to develop an ad hoc SQL query and submit it e.g. R and RStudio
- Tools, which aim to automatically develop SQL queries based on user's input (BI software)

Summary

- SQL is a key language in databases domain
- The knowledge of SQL is fundamental for:
  - Software developers
  - Data scientists
  - Database administrators
- In spite of standardisation efforts, different SQL dialects exist
- While most of SQL knowledge can be transferred between DBMSs, SQL queries are typically not portable due to dialect differences

Pytania ?

Zapraszam w trakcie wykładu
i w trakcie konsultacji

Miejsce i termin konsultacji
- na stronie:
https://pages.mini.pw.edu.pl/~grzendam/pl/dydaktyka.html

Questions?

I am looking forward to your questions during the lectures (including now) and during my office hours

For details on my office hours, please check my website:

https://pages.mini.pw.edu.pl/~grzendam/eng/dydaktyka_eng.html

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